

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct:

I P Surendra Kumar Reddy (192425224) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P Surendra Kumar Reddy

CONCEPT MAPPING

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.

Analyze the missing layers and explain how their functions are essential for reliable communication.

Answer:

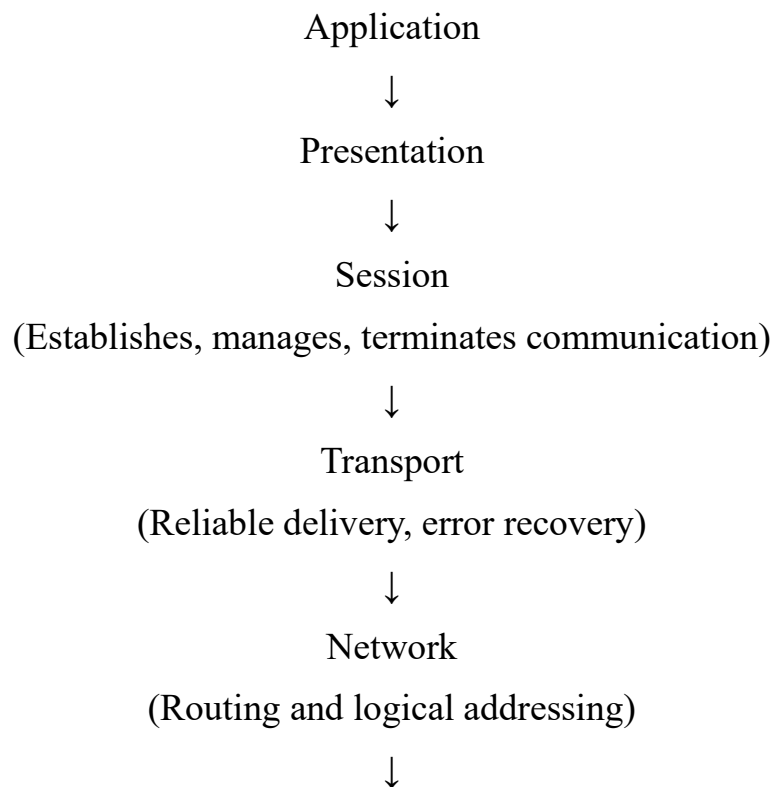
Missing Layers:

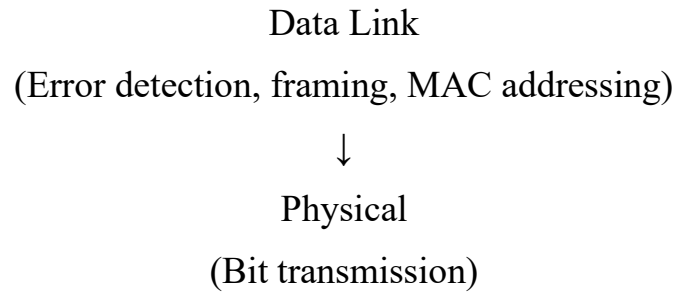
- **Session Layer**
- **Data Link Layer**

Correct OSI Layers:

Application → Presentation → **Session** → Transport → Network → **Data Link** → Physical

Concept Map



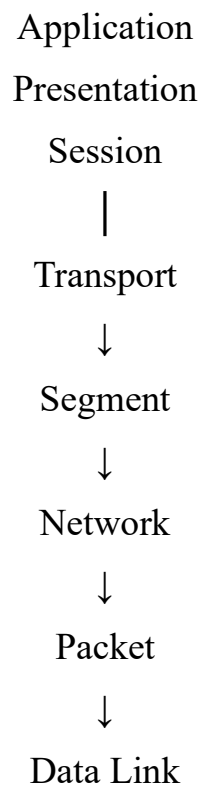


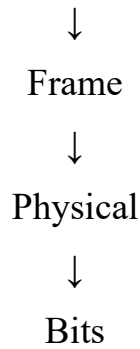
Analysis

- **Session Layer** creates and maintains communication sessions between devices.
- **Data Link Layer** detects transmission errors and delivers frames within the local network.
- Together they improve reliability by maintaining communication and ensuring error-free frame delivery.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

Concept Map





Analysis

- Application, Presentation and Session generate user data.
- Transport converts data into **Segments**.
- Network converts segments into **Packets**.
- Data Link converts packets into **Frames**.
- Physical converts frames into **Bits** for transmission.
- At the receiver, the process is reversed (Decapsulation), ensuring successful communication.

3. Given the concept map:

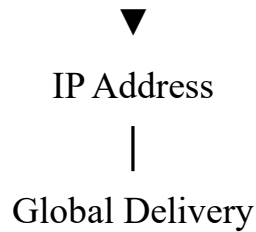
DataLink → MACAddress → LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

Concept Map





Analysis

- **MAC Address** uniquely identifies devices within a local network.
- **IP Address** identifies devices across different networks.
- Data Link Layer handles communication inside a LAN.
- Network Layer routes packets between different networks.
- Together they enable complete end-to-end communication.

4. A concept map shows:

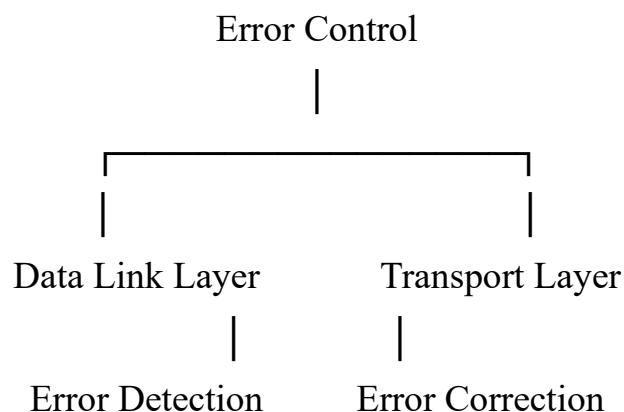
ErrorControl

→DataLinkLayer→ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

Concept Map

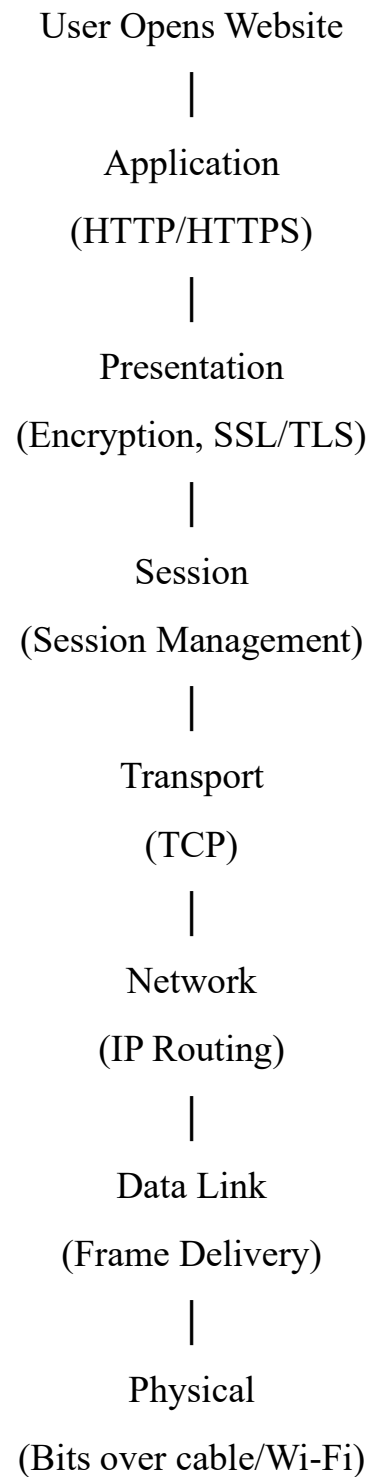


Analysis

- Data Link Layer detects frame errors using CRC or checksum.
- Transport Layer ensures reliable delivery using acknowledgements, retransmission and flow control.
- The combination provides accurate and reliable communication.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

Concept Map



Analysis

- Application requests the webpage.

- Presentation encrypts and formats data.
- Session manages the browsing session.
- Transport provides reliable data transfer.
- Network routes packets.
- Data Link delivers frames.
- Physical transmits bits.

If one layer fails:

- Physical fails → No signal.
- Data Link fails → Frames lost.
- Network fails → No routing.
- Transport fails → TCP connection breaks.
- Application fails → Webpage cannot load.

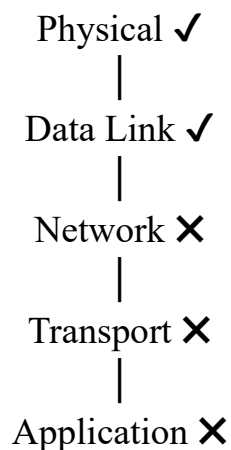
Thus, every layer depends on the previous one

6. A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✗ → Transport ✗ → Application ✗

Analyze the problem and explain how the concept map helps in troubleshooting.

Given Concept Map



Analysis

- Physical Layer is functioning properly.
- Data Link Layer is working correctly.

- Network Layer has failed (routing/IP issue).
- Since packets cannot be routed, Transport Layer cannot establish communication.
- Consequently, the Application Layer cannot access services like websites or email.

Possible Causes

- Incorrect IP address
- Router failure
- Wrong subnet mask
- Missing default gateway
- Routing table errors

Troubleshooting Steps

1. Check IP configuration.
2. Verify subnet mask and default gateway.
3. Ping the local gateway.
4. Check router status.
5. Verify routing tables.
6. Restart network devices if necessary.

Conclusion:

The concept map clearly identifies that the failure starts at the **Network Layer**, making troubleshooting faster by narrowing down the source of the problem.